

ORIGINAL ARTICLE

Structural Evaluation of *RYR2*-CPVT Missense Variants and Continuous Bayesian Estimates of Their Penetrance

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BACKGROUND: Catecholaminergic polymorphic ventricular tachycardia (CPVT) is strongly associated with rare missense variants in *RYR2*, the gene encoding the intracellular calcium release channel RyR2.

METHODS: We curated 179 articles reviewed by 3 individuals to extrapolate *RYR2*-CPVT missense variant genotype-phenotype relationships. Purportedly neutral control variants were ascertained from *RYR2* missense variants observed in gnomAD and ClinVar. We performed an *RYR2*-CPVT Bayesian penetrance analysis by conditioning a CPVT penetrance prior (hereafter called Bayesian CPVT-prior) on variant-specific features (in silico and structural) calibrated by heterozygote phenotypes. We evaluated various features, including the Bayesian CPVT-prior, ClinVar, REVEL, and AlphaMissense¹⁰ against our observed CPVT penetrance, using Spearman rank-order correlations and Brier Scores. Penetrance estimates were superimposed on a cryo-EM structure of RyR2 to investigate hot-spot heterogeneity.

RESULTS: From the literature, we identified 1014 affected missense *RYR2* heterozygotes (468 unique variants), and further supplemented by purportedly unaffected heterozygotes in GnomAD to a total of 622575 heterozygotes (5181 unique variants). Among the predictors, the Bayesian CPVT-prior score had the highest Spearman rank-order and lowest Brier scores, respectively (0.19; 0.0090), compared with ClinVar (0.083; 0.019), REVEL (0.16; 0.018), or AlphaMissense (0.18; 0.018). Penetrance estimates for all *RYR2* missense variants are prospectively hosted at the Variant Browser website.

CONCLUSIONS: Traditional categorical variant annotations limit accurate disease risk assessment, especially in populations unascertained for disease. A probabilistic, population-calibrated penetrance framework enables more meaningful prospective variant interpretation. Our Bayesian CPVT-prior outperforms current tools in evaluating *RYR2* variant penetrance. We provide prospective Bayesian CPVT penetrance values for 29242 *RYR2* missense variants at our online variant browser.

Key Words: calcium ■ counselors ■ heterozygotes ■ penetrance ■ sarcoplasmic reticulum

Inherited arrhythmia syndromes commonly arise from variants in genes encoding cardiac ion channels, their modulating proteins, and cardiomyocyte structural proteins.^{1–3} Catecholaminergic polymorphic ventricular tachycardia (CPVT) is 1 such rare syndrome resulting in atrial and ventricular arrhythmias, most

frequently during periods of high adrenergic stimulus. Studies of the genetic basis of CPVT have revealed both autosomal dominant and autosomal recessive patterns of inheritance. Recently, the ClinGen variant curation working group identified strong evidence for the involvement of the following genes in CPVT: *RYR2*,

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Nonstandard Abbreviations and Acronyms

ACMG	American College of Medical Genetics and Genomics
AUC	area under the curve
CPVT	catecholaminergic polymorphic ventricular tachycardia
CRDS	calcium release deficiency syndrome
EM	expectation-maximization
P/LP	pathogenic/likely pathogenic
VUS	variant of uncertain significance

CASQ2, *TECL*, *TRDN*, and *CALM1-3* (Figure 1A).³ *RYR2* encodes for a calcium-activated calcium-release channel located intracellularly in the sarcoplasmic reticulum, the major Ca²⁺ storage organelle in cardiac muscle. During excitation-contraction coupling, RyR2 is activated by Ca²⁺ influx from voltage-gated L-type Ca²⁺ channels located in the cell membrane, a process known as Ca-induced Ca-release. Autosomal dominant *RYR2* gain-of-function variants predispose to spontaneous channel openings and sarcoplasmic reticulum Ca²⁺ release during diastole, resulting in ventricular ectopy and triggers of reentrant ventricular arrhythmias. Unfortunately, these events can result in sudden cardiac death and may be the first manifestation of an otherwise unrecognized disease.

Early identification of *RYR2*-CPVT variant heterozygotes may help prevent sudden cardiac death if appropriate treatment is initiated. Traditionally, genetic evaluation has been accomplished after an *RYR2*-CPVT proband is identified, followed by variant-specific familial cascade screening. An alternative strategy would be to use genotype-first approaches to preemptively identify genetically at-risk individuals before the emergence of events.^{4,5} Progress towards this goal has been stymied by the frequency of genetic noise occurring in this large protein (15 kb), the lack of high-throughput functional approaches to characterize variants, and the rarity of this syndrome (reported incidence of 1:10 000).⁶ The American College of Medical Genetics and Genomics (ACMG) has developed a framework of variant classification to guide clinician management of genetic conditions, relying on functional, clinical, in silico, experimental/functional, and other evidence criteria to classify variants.⁷ Classifications span from benign/likely benign to pathogenic/likely pathogenic (P/LP), depending on various evidence strengths. Unfortunately, most variants lack sufficient information for classification and instead fall under the nebulous category of variants of uncertain significance (VUS). These variants are not actionable clinically and introduce confusion and anxiety among heterozygote individuals. This is especially problematic for *RYR2* variants, given that 3912 of 7783 single-nucleotide variants

in the clinical genomics resource ClinVar are currently classified as VUS.⁸

Recently, *RYR2* loss-of-function variants have been linked to a distinct genetic sudden death syndrome known as calcium release deficiency syndrome (CRDS),⁹ with clinical features and mechanistic foundations that differ from CPVT caused by *RyR2* gain-of-function disease.¹⁰ Because CRDS results from reduced RyR2 channel function rather than adrenergically triggered spontaneous diastolic sarcoplasmic reticulum Ca²⁺ release, the clinical phenotype and optimal testing strategy differ from CPVT.¹¹ For this reason, loss-of-function or CRDS variants require separate consideration from CPVT gain-of-function variants, and penetrance estimates derived for CPVT are not directly applicable to proven CRDS variants. We explicitly flagged published CRDS variants in our Variant Browser and treat them as not applicable for CPVT penetrance interpretation (see methods and Table S1).

As a higher resolution alternative to ACMG classification, we pioneered a Bayesian penetrance–effect size framework that converts variant-specific data into estimates of disease penetrance.^{12–15} This approach addresses 2 conceptual limitations of strict categorical ACMG labels. First, binary designations, such as benign or likely benign and P/LP, do not capture intermediate levels of risk. In rare circumstances, a variant with limited supporting evidence may carry low but nonzero risk when combined with additional genetic or environmental factors, or when insufficient evidence has led to misclassification. Second, many heterozygotes for P/LP variants will never exhibit symptoms, which reflects the problem of incomplete penetrance.¹⁴ Our probabilistic penetrance framework better represents graded disease risk. Consistent with the ACMG system, we incorporate multiple lines of evidence (structural data, in silico, experimental, where available); however, unlike the ACMG system, our approach is conditioned on prior probabilities from extensive variant curation with quantification of estimated uncertainty.

In this study, we define penetrance as the probability that a heterozygous carrier of an *RYR2* variant will manifest a CPVT phenotype by the time of observation. Our inherently probabilistic approach complements typical clinical reasoning, where diagnostic and treatment decisions are based on forecasting event probabilities (here, syncope, ventricular arrhythmias, sudden cardiac death). Our group has previously deployed this strategy to study other arrhythmia syndromes, including Brugada and long-QT syndrome 1–3.^{12,14,15} In addition to providing penetrance estimates, we have carefully curated variant in silico, experimental, and population evidence in a conveniently hosted Variant Browser (variantbrowser.org) to facilitate efforts of physicians, genetic counselors, and researchers to collectively advance the care of at-risk individuals.

Accordingly, we apply this strategy to provide comprehensive estimates of penetrance for *RYR2*-CPVT

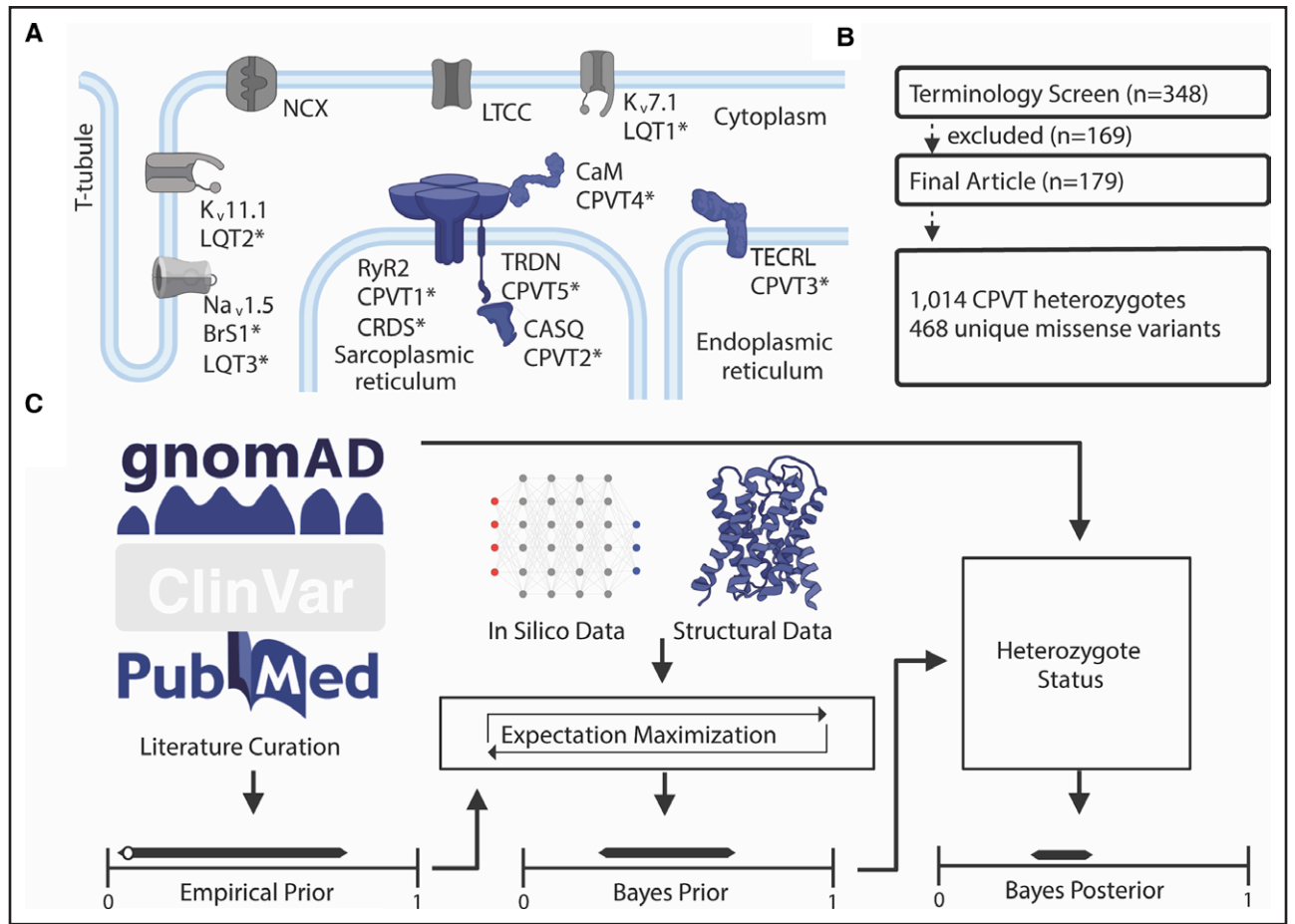


Figure 1. Schematic of catecholaminergic polymorphic ventricular tachycardia (CPVT)-associated proteins and workflow of literature curation and model implementation.

A, Cellular schematic depicting key cardiomyocyte ion channels. These include ion channels associated with other channelopathies, Nav1.5, Kv11.1, and Kv7.1, as well as supporting proteins. Genes are denoted with an asterisk (*). LQT (long QT), Brugada syndrome (BrS), CPVT, calcium release deficiency syndrome (CRDS), NCX (sodium-calcium exchanger), LTCC (L-type calcium channel), CaM (calmodulin), CASQ (calsequestrin), TECRL (trans-2,3-enoyl-CoA reductase-like), TRDN (triadin). **B**, Flowchart depicting literature curation. We first searched PubMed for the terms CPVT and RYR2, which returned 348 articles. On further analysis, 179 reported variant data with sufficient information to detect an affected vs unaffected phenotype. These resulted in a total of 1014 CPVT heterozygotes harboring 468 unique missense variants. Suspected duplicates were removed when multiple reports from the same institution repeatedly reported on a unique variant. Further analyses were supplemented by GnomAD purported unaffected individuals as described in the article. **C**, Graphical schematic of Bayesian penetrance model. Penetrance is defined as the fraction of affected vs total heterozygotes harboring a particular variant. Here, we collect variant observational data from ascertained and unascertained sources and deploy a model to integrate these data with variant-specific features (in silico and structural data). Observationally, most rare variants have an observed, or empirical, penetrance of 0% or 100%—our current approach aims to provide a more refined estimation of penetrance for rare variants without tens to hundreds of observations.

missense variants. Our approach relies on extensive curation of the literature, population databases, and structural patterns in RyR2. We compare the value of our penetrance estimates and structural calculations to contemporary tools AlphaMissense,¹⁶ REVEL,¹⁷ and existing ClinVar⁸ classifications to highlight interesting insights into structural features revealed through these estimates. All data are provided at the Variant Browser (variantbrowser.org) for community uptake and use.

METHODS

All data and materials have been made publicly available on the Variant Browser and can be accessed at variantbrowser.org.

All code to process these results is available at <https://github.com/kroncke-lab/>.

Institutional review board (no. 191563) of Vanderbilt University Medical Center granted a waiver for this work based on 45 CFR 46.104 (d) category (4) for Exempt Review.

Curation of Heterozygote Variant Data

We performed a PubMed literature search with the terms RYR2 and CPVT, which resulted in 348 articles on June 29, 2022. Based on their abstracts, 179 articles underwent additional analysis. Three independent reviewers assessed each article for an RYR2 variant and affected heterozygote status, as ascertained by the authors of each article. For individual variants, we avoided repeated heterozygote counting by excluding articles

that reported duplicate patients reported by each institution; each study was independently cross-checked by searching for articles detailing the same variant and phenotype from the same authors/institution. We used the *RYR2* transcript ENST00000366574 throughout. We gathered unaffected *RYR2*-CPVT heterozygotes from gnomAD v4.1.0. We have previously reported sensitivity analyses for more prevalent inherited arrhythmia syndrome genotype-phenotypes, showing that this assumption does not over-inflate downstream analyses.¹⁵ The literature curation and gnomAD data set were summed by variant, and the count of affected and unaffected status was accordingly assigned (Table S1; Figure 1B). Compound heterozygotes were excluded from this study.

We separately interrogated the literature for purported loss-of-function variants associated with CRDS (Table S1). We identified 12 missense variants and 1 in-frame deletion that had evidence to support CRDS, harbored by a total of 39 heterozygotes. Given their different underlying biology, we identified these as not applying to CPVT penetrance.

Penetrance Calculations

Using a previously reported strategy, we implemented Bayesian estimates of penetrance and their uncertainty using shared information across all *RYR2*-CPVT variants.¹² Formally, the observed penetrance was calculated as the fraction of affected (α) over the sum of affected (α) and unaffected (β) heterozygotes for each *RYR2* variant. The rare incidence of CPVT precluded accurate estimation due to sample size and ascertainment bias, but could approximate 'truth' in larger data sets. For each variant i , α , and β reflected the number of affected and unaffected individuals heterozygous for the variant, respectively.

$$\text{Observed Penetrance}_i = \frac{\alpha_i}{\alpha_i + \beta_i} \quad (1)$$

In this study, penetrance was defined as the probability that an *RYR2* variant heterozygote exhibits a CPVT phenotype by the time of observation. A heterozygote was classified as affected if the original report documented (1) a clinical diagnosis of CPVT, (2) exertion-associated ventricular ectopy or exercise-induced polymorphic ventricular tachycardia consistent with CPVT, (3) syncope, seizure, aborted sudden cardiac arrest, or sudden cardiac death attributed to CPVT or a compatible arrhythmic event, or (4) a positive exercise test interpreted as consistent with CPVT. Individuals explicitly reported as asymptomatic after comprehensive evaluation, including negative exercise testing when performed, were considered unaffected. Reports lacking sufficient detail to determine phenotype were excluded. Because thresholds for defining a positive exercise test vary across studies, we relied on the interpretation of the reporting authors. This approach introduces heterogeneity but reflects the constraints of retrospective literature curation. A prospective analysis with standardized exercise-testing criteria would provide more uniform penetrance estimates, but it was not possible here.

CPVT Empirical Prior

The 2 parameters, $\alpha_{\text{prior, empirical}}$ and $\beta_{\text{prior, empirical}}$ were estimated for all *RYR2* variants using a beta binomial model with weighted average (w_i) (see equation 6 below) and mean squared error across all variants (implementation in GitHub). Together, the parameters defined a unique Empirical Prior applied to all *RYR2*

variants. In the absence of additional variant-specific information, this value approximated the uniform penetrance of all *RYR2*-CPVT variants.

$$\text{Empirical Prior} = \frac{\alpha_{\text{prior, empirical}}}{\alpha_{\text{prior, empirical}} + \beta_{\text{prior, empirical}}} = \sum_i \frac{\alpha_i}{\alpha_i + \beta_i} w_i \quad (2)$$

CPVT Empirical Posterior

We refined the CPVT Empirical Prior using individual variant (i) observed affected (α_i) and unaffected (β_i) heterozygotes as a likelihood. This delivered a best estimate of penetrance for each variant based on observation alone.

$$\text{Empirical Posterior}_i = \frac{\alpha_i + \alpha_{\text{prior, empirical}}}{\alpha_i + \beta_i + \alpha_{\text{prior, empirical}} + \beta_{\text{prior, empirical}}} \quad (3)$$

EM Algorithm

The empirical posterior enabled the derivation of point estimates of a Bayesian prior and estimated uncertainty. The *RYR2*-CPVT penetrance estimates were generated using a previously described expectation-maximization (EM) approach.¹² Regression used variant-specific features as independent covariates, and the empirical posteriors of those variants as dependent variables. Multiple rounds of EM were performed until iterative convergence of each probability point estimate was achieved, as informed through $\alpha_{\text{prior, empirical}}$ and $\beta_{\text{prior, empirical}}$. This involved 3 unique steps: (1) use an empirical Bayes penetrance model to derive penetrance scores; (2) penetrance estimate regression to variant features; (3) point estimate revisions to convergence.

Bayesian Prior

Individual variant point estimates and uncertainty were defined using the new $\alpha_{\text{prior, EM}, i}$ and $\beta_{\text{prior, EM}, i}$.

$$\text{Bayesian Prior}_i = \frac{\alpha_{\text{prior, EM}, i}}{\alpha_{\text{prior, EM}, i} + \beta_{\text{prior, EM}, i}} \quad (4)$$

Bayesian Posterior

If clinical data were available for the variant, then the observations (α and β in equation 1) were applied as another likelihood

$$\text{Bayesian Posterior}_i = \frac{\alpha_i + \alpha_{\text{prior, EM}, i}}{\alpha_i + \beta_i + \alpha_{\text{prior, EM}, i} + \beta_{\text{prior, EM}, i}} \quad (5)$$

The most clinically interesting variants are private mutations. To avoid overweighting of known benign or CPVT-associated variants, as well as in penetrance estimates calculated above, we used the following weighting for each variant i :

$$1 - \frac{1}{0.01 + \text{total heterozygotes}_i} \quad (6)$$

RYR2 Variant In Silico, Clinical, and Structural Features

We obtained scores for all *RYR2* missense variants from AlphaMissense¹⁶ and REVEL,¹⁷ as previously described. *RYR2* variant classifications were obtained from the ClinVar resource on 5/11/2024.⁸ Following current clinical practice, VUS were considered unaffected along with benign/likely benign variants, whereas P/LP were considered affected when performing model evaluations. We analyzed the cryo-EM structure of a closed RyR2 (PDB: 7U9Q).¹⁸ We calculated the CPVT-density feature as a structural prediction based on the 3-dimensional-environment of each residue across RyR2, following previous

implementations.¹⁹ Regions and hot-spots of the protein are detailed in Table S2. When comparatively evaluating these metrics, we used a Leave One Out Cross Validation strategy to avoid over-estimation of performance. For prospective estimates, we leveraged the entirety of the available data.

Statistical Analysis

To determine covariate performance on a binary outcome of affected/unaffected status, we used 2 analyses: (1) area under the curve (AUC) for a receiver-operator curve and (2) AUC for a Precision-Recall curve. These analyses were executed using R version 4.1.2, with the packages pROC and PRROC. Spearman rank order coefficients were determined in R using the package wCorr. We performed 5-fold cross-validation using the Caret package in R. The data set was randomly divided into 5 equal-sized folds. In each iteration, 1-fold was used as the test set, while the remaining 4-folds were used for training. This process was repeated 5x, ensuring that each fold was used as a test set once. To evaluate the model's performance, we calculated the weighted Spearman correlation between the posterior and empirical penetrance estimates. Probabilistic Brier scores were used to evaluate covariate forecasting as defined below²⁰:

$$\text{Brier Score} = \frac{1}{N} \sum_{t=1}^n (f_t - o_t)^2 \quad (7)$$

N is the quantity of observed variants, f_t is the estimated penetrance for the t^{th} variant, and o_t is the observed penetrance of the t^{th} variant (Equation 3). Brier scores were calculated using different filtering allele frequencies as specified in the text. All data and code necessary to perform the analyses are publicly available at <https://variantbrowser.org/RyR2/> and <https://github.com/kroncke-lab/RyR2-disease-penetrance>.

RESULTS

Curation of a Set of Affected/Unaffected Heterozygotes With CPVT

To develop a comprehensive set of affected and unaffected *RYR2* heterozygotes, we undertook a comprehensive literature curation of reports detailing genetic testing and clinical presentations of individuals undergoing clinical investigation (see methods; Figure 1B). An in-depth manual review of the articles identified 179 publications detailing 544 *RYR2* variants related to CPVT. Among variant heterozygotes in these reports, 1014 individuals were categorized as affected and 375 as unaffected based on the reporting authors' clinical assessments (see methods for precise phenotype criteria). Affected status was not limited to the occurrence of cardiac arrest; it also included exertion-associated ventricular ectopy, positive exercise testing interpreted as consistent with CPVT, and author-reported clinical CPVT diagnoses. We separately interrogated the literature for purported loss-of-function variants associated with CRDS (Table S1). We interrogated gnomAD for purportedly unaffected *RYR2* missense variant heterozygotes. Filtering on an allele frequency of <0.01%,

we found a total of 5159 variants harbored by 34396 individuals. Based on previous sensitivity analyses for long-QT syndrome and the comparatively lower incidence of CPVT, we do not expect our conclusions to change based on the assumption of unaffected status.^{12,15}

Results of CPVT-Density and CPVT Bayesian Penetrance Calculations

Using the curated set of *RYR2* variants, we first calculated CPVT-density, a structural metric based on the 3-dimensional proximity of affected CPVT *RYR2* missense variants across the protein structure. We then determined Bayesian CPVT-prior and posterior scores for variants as described in the methods (Figure 1C). These findings are hosted on the variant browser (variantbrowser.org). Figure 2A shows the distributions of observed penetrance and Bayesian penetrance. Most variants tended to have an observed penetrance of 0 or 1, reflecting bias from population databases as well as small numbers of heterozygotes studied in case reports. In contrast, there was a comparative smoothing of the distribution with the Bayesian CPVT-prior—no variants with estimated complete penetrance and very few with estimated complete absence of disease. Figure 2B highlights focal areas of RyR2 with enhanced penetrance. Hot-spot criteria have previously been applied in the ACMG framework to facilitate variant classification for inherited arrhythmia syndromes.²¹

Using a published cryo-EM structure of the closed RyR2 channel,¹⁸ we superimposed penetrance estimates across the protein structure (Figure 3). We observed many variants with low penetrance in traditional hot-spot domains: V186M (1/23 affected); in BsolB: R2258C (1/19 affected); in Csol: I3995V (1/8 affected); in Carboxy Terminal Domain: R4959W (1/44 affected); in TMD: A4723T (1/24 affected). Interestingly, we observed moderate penetrance among recurrently reported CPVT-associated variants, such as R420W in the NTD (14/24 affected). Using our Bayesian penetrance observations, segments where average penetrance density exceeded >0.4 across a 100 amino acid span with >3 variants, only 5 areas met these criteria—residues 165 to 189 (8 variants), 401 to 415 (4 variants), 2082 to 2484 (48 variants), 3876 to 4204 (55 variants), 4488 to 4959 (54 variants; Figure 3). These data demonstrate the complexity beyond the hot-spot categorized by domain, especially when visualized on the protein structure (Figure 3).

Evaluation of Bayesian Penetrance, CPVT-Density, In Silico, and Clinical Predictors

Next, we evaluated the association of the Bayesian CPVT-prior, CPVT-density, ClinVar, and the in silico predictors AlphaMissense and REVEL against the Empirical

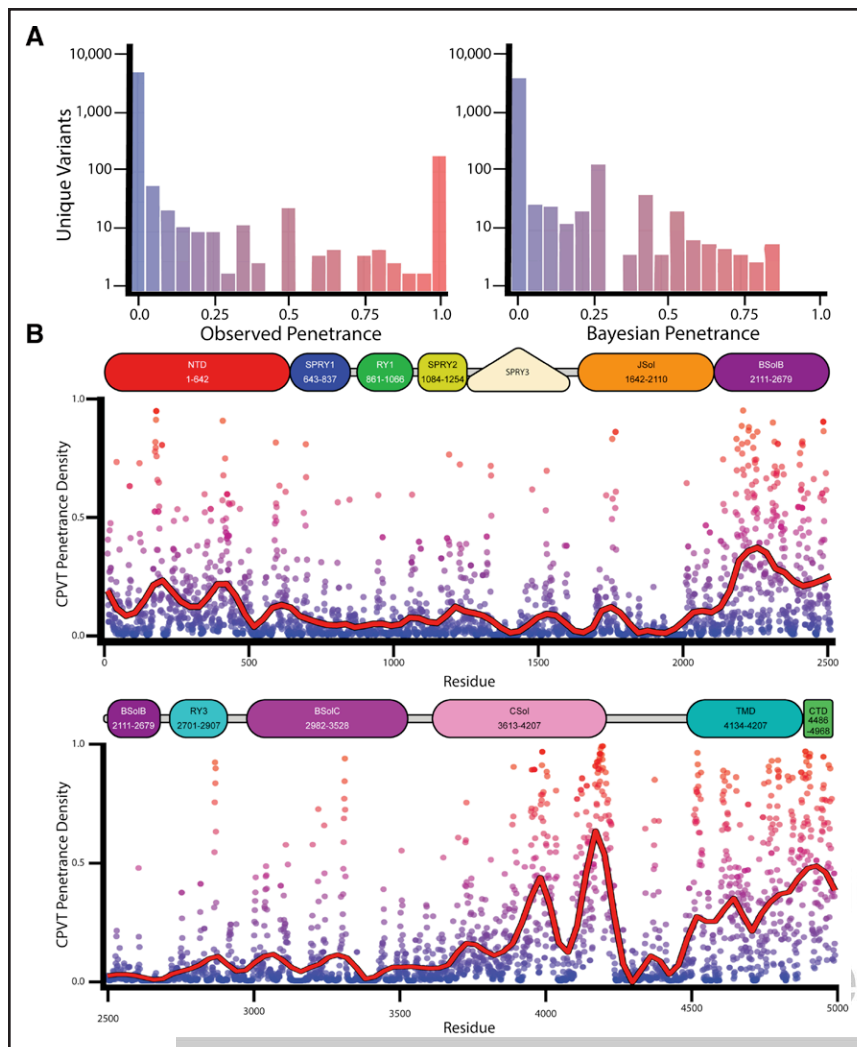


Figure 2. Prior mean catecholaminergic polymorphic ventricular tachycardia (CPVT) penetrance is in high agreement with RYR2 topology.

A. Histograms of *RYR2* variant penetrance counts between the observed and Bayesian penetrance models (y axis of counts, logarithmic scale). In the Bayesian analysis, comparatively fewer variants have a predicted near-complete penetrance, vs the observed penetrance from limited sampling. Missense variants of *RYR2*-CPVT are reported with each point. Color indicates the prior mean CPVT penetrance on a red-blue spectrum (1–0, respectively). B. Plots of residue-averaged penetrance across the length of *RYR2*. Residue numbers are reported alongside RyR2 schematic domains. Traditional hot-spots have higher average penetrant residues (NTD [N-terminal domain], BSolB [bridge solenoid], BSolC, CSol (central solenoid), TMD [transmembrane domain], and CTD [C-terminal domain]; 1–639, 2109–3554, 3634–4130, 4490–4886, 4887–4914, respectively). NTD, SPRY (SP1a and RYanodine receptor domain), RY (RyR repeat), JSol (junctional solenoid), BSol, CSol, TMD, and CTD.

Posterior (Figure 4A; Table 1). We found that all covariates had nominally significant ($P < 0.05$) associations. We derived the highest Spearman rank-order correlation coefficient with the Bayesian Prior (0.193), with the lowest performance by ClinVar scores (0.083). Interestingly, AlphaMissense (0.182) and REVEL (0.156) had higher coefficients than that of the CPVT density score (0.146). The above analysis is not perfectly suited, as incomplete penetrance is an underlying stochastic process (a probability of disease given the presence of a variant). Accordingly, we next calculated Brier Scores for each predictor as a complementary, probabilistic evaluation metric (lower scores indicate improved performance; Table 2). We found that for Bayesian CPVT, penetrance had the lowest Brier score (0.0090) compared with CPVT penetrance density (0.018), REVEL (0.018), AlphaMissense (0.018), or ClinVar (0.019).

Finally, we determined receiver operating characteristic curves for each model covariate (Figure 4B). The resulting AUCs from logistic models trained to predict *RYR2* were as follows (AUC; [95% CI]): REVEL (0.87; [0.81–0.93]), AlphaMissense (0.75; [0.68–0.82]), CPVT

penetrance density (0.82; [0.77–0.99]), Bayes prior (0.92; [0.84–0.99]), ClinVar (0.63 [0.51–0.75]). We show covariate AUCs at various cutoff thresholds in Figure S1. As penetrance was not equally distributed across variants but more heavily weighted towards zero, we also assessed Precision-Recall AUC to address the imbalanced data set (Figure 4C). The prior mean predictor performed well (0.28 [0.23–0.33]). CPVT-density (0.26 [0.21–0.31]) and REVEL (0.12 [0.072–0.17]) showed moderate performance, while AlphaMissense (0.090 [0.040–0.14]) and ClinVar (0.047 [–0.0030 to +0.097]) performed poorly.

DISCUSSION

RYR2-CPVT Penetrance Estimates Improve Variant Interpretation. Here, we develop a predictive model of *RYR2*-CPVT missense variant penetrance to facilitate structural understanding and clinical evaluation of this lethal arrhythmia syndrome. Variant penetrance is increasingly emerging as a critical component of clinical genetics, particularly for arrhythmias and cardiomyopathies.^{22,23}

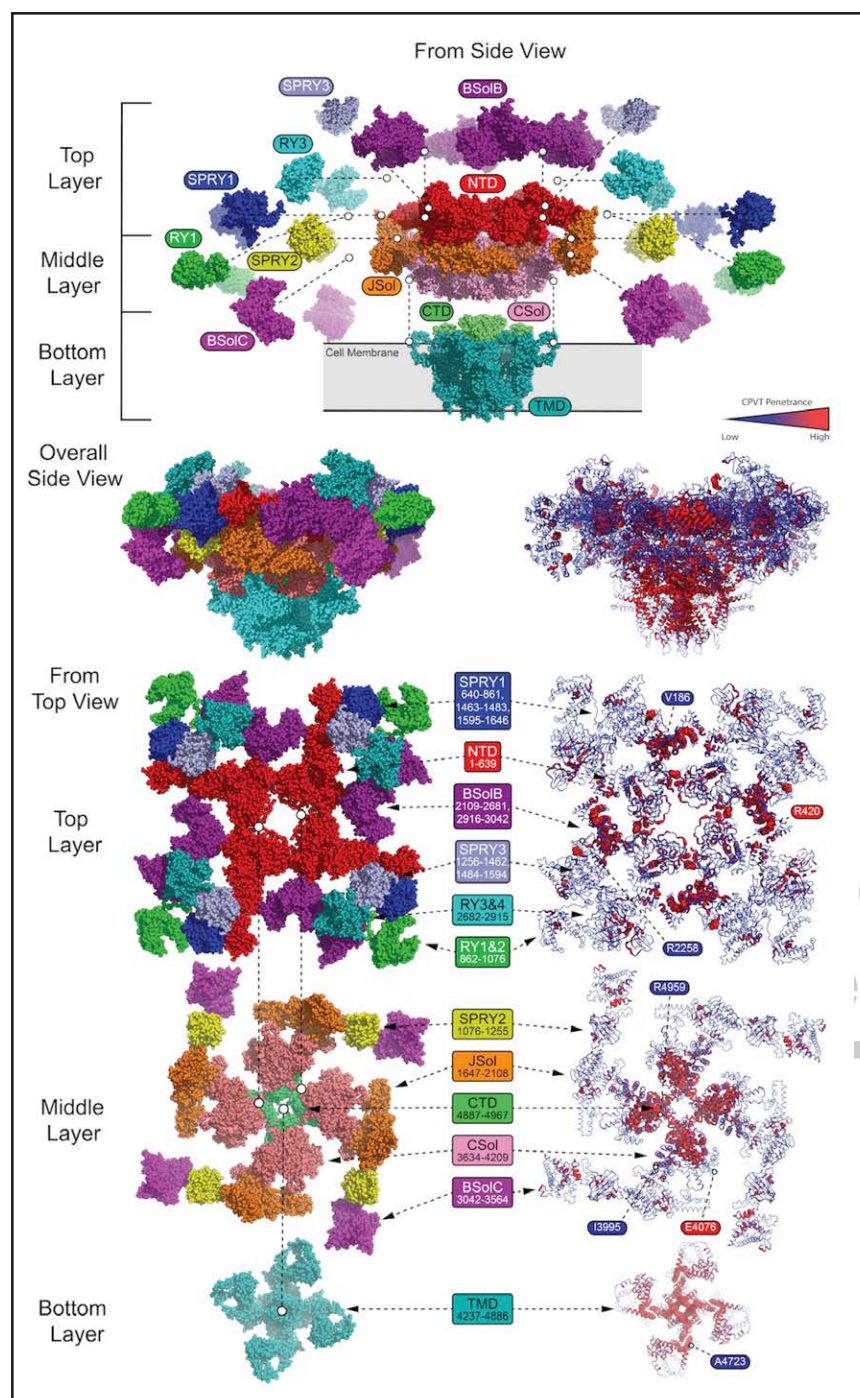


Figure 3. Bayesian penetrance estimates mapped onto structure.

Cryo-EM structure of RyR2 (PDB: 7U9Q).¹⁵ **Left** depicts previously defined structural regions, **right** reflects penetrance estimates superimposed on structure, averaged over each residue. Traditional hot-spots show heterogeneity of penetrance, such as BSolB. NTD (N-terminal domain), SPRY (SP1a and ryanodine receptor domain), RY (RyR repeat), JSol (junctional solenoid), BSol (bridge solenoid), CSol (central solenoid), TMD (transmembrane domain), CTD (C-terminal domain).

Whereas variant classification may help define the variant's etiological contribution to disease (ie, ACMG), penetrance assesses the more difficult, prospective question of whether variant heterozygotes will manifest a certain disease. Unfortunately, many studies conflate these questions, using ClinVar classifications as proxies for variant penetrance and severity. We and others have shown that ACMG variant classifications are often poor markers of disease penetrance.^{14,24} Moreover, variant *prognosis* can query more granular levels of such phenotypes (age of onset, severity of manifestations, and recurrence

of manifestations). Our Bayesian approach addresses both latter questions in previous work on *KCNH2*-LQT2, using a richly phenotyped clinical data set.¹³ In the present study, we are limited to penetrance assessment due to the currently less deeply phenotyped nature of our clinical data sets; however, we show that our Bayesian approach outperforms current variant effect predictors and clinical interpretations of ClinVar classifications in addressing penetrance. Our approach is fundamentally phenotype-agnostic but has been primarily deployed to study inherited arrhythmia syndromes.^{12,14,15} Deploying

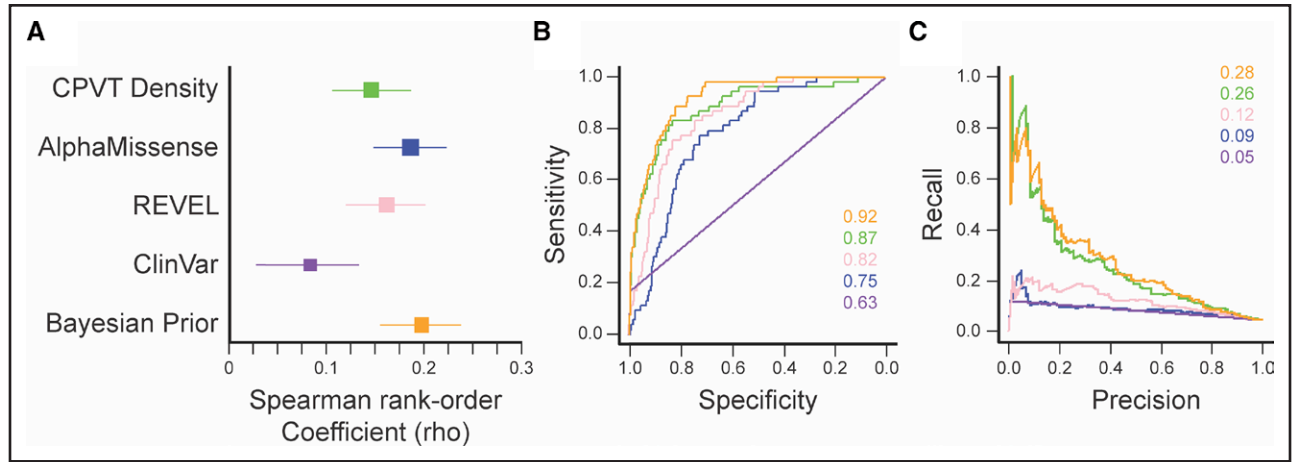


Figure 4. Catecholaminergic polymorphic ventricular tachycardia (CPVT) Bayesian model covariate evaluation and penetrance estimates.
A, Forest Plot of Spearman correlations among model covariates. Optimal performance was obtained with a Bayesian prior. **B**, Receiver Operating Characteristic plot of model covariates. Areas under the curve are displayed in the inset. Colors correspond to those in **A**. **C**, Precision-Recall curve of each covariate. Areas under the curve are displayed in the inset. Colors correspond to those in **A**. Higher scores indicate better performance.

our method on additional genes within cardiomyopathies and arrhythmias will complement previous disease (GENESIS)²⁵ and tissue-specific (CVD-PP²⁶) in this space. Moreover, the combination of a close literature review, previous cohort studies, and a population database complements recent penetrance frameworks specifically aimed at the secondary finding population.²⁷

Clinical Application of CPVT Penetrance Estimates

Genetic testing is increasingly incorporated into clinical diagnostic frameworks, including the Schwartz Score for long-QT syndrome²⁸ and the Shanghai Score for Brugada Syndrome.²⁹ As larger genomic data sets become available, with a heightened understanding of underlying architecture, these genetic data would be expected to carry more weight. Smoothly integrating clinical observations with genetic data promises to enhance diagnosis and prognosis. In 1 such application, a CPVT clinical score was developed and externally validated, which insightfully combined clinical and genetic data to facilitate phenotype-enhanced variant classification.³⁰

Accordingly, phenotype-enhanced variant classification, defined as interpreting genetic variants in the context of focused clinical testing such as exercise stress testing, has been successfully used to reclassify *RYR2* VUS. Because such focused testing may be performed once a candidate variant is identified, clinicians can often resolve the relevance of an *RYR2* VUS at the individual level. In contrast, genotype-first workflows that identify thousands of heterozygotes with many distinct variants will require alternative approaches. Our Bayesian penetrance estimates, therefore, may serve as a prioritization and triage tool to identify variant heterozygotes most warranting targeted phenotyping after ascertaining on genotype alone in secondary findings.

Accordingly, the current method may facilitate the application of proactive genotype-first approaches. As larger groups of individuals undergo whole genome sequencing, prospectively available penetrance estimates will identify at-risk individuals who would benefit from preemptive clinical workup, including exercise stress tests.⁶ In such a case, the prior would be based on variant data, with physical exam and diagnostic testing providing a likelihood regarding disease status. A

Table 1. Spearman Correlation Coefficients by CPVT Model Covariates

Covariate	Allele frequency <1%	All
Bayesian CPVT penetrance	0.585	0.589
CPVT penetrance density	0.146	0.146
AlphaMissense	0.182	0.186
REVEL	0.156	0.162
ClinVar annotation	0.083	0.083
Prior mean	0.193	0.198

CPVT indicates catecholaminergic polymorphic ventricular tachycardia.

Table 2. Brier Scores for Model Covariates Stratified by Variant Frequency

Covariate	Allele frequency <1%	All
Bayesian CPVT penetrance	0.0090	0.00081
CPVT penetrance density	0.018	0.0015
AlphaMissense	0.018	0.0016
REVEL	0.018	0.0015
ClinVar annotation	0.019	0.0016
Prior mean	0.017	0.0015

CPVT indicates catecholaminergic polymorphic ventricular tachycardia.

current challenge remains the significant incomplete penetrance associated with inherited arrhythmia syndromes. For example, an early exome sequencing study recognized the incomplete penetrance of reported *RYR2*-CPVT variants, which would have resulted in a prevalence of 1:150, a study that preceded the current ACMG guidelines.^{7,31} When applied correctly, our approach could potentially decrease morbidity and mortality by early initiation of treatment, such as beta blockers and lifestyle modification, before manifestations of overt ventricular arrhythmias.³² This question is especially important given that *CPVT* variants are now reported as secondary findings.³² In a recent arrhythmia-focused biobank study, the actionability of discovered *RYR2* variants was limited by the lack of P/LP annotations.⁵ This challenge could be overcome by the current approach of focusing on readily available penetrance estimates. Moreover, high-throughput experimental studies of inherited arrhythmia syndrome-associated genes, such as *RYR2*, further promise to refine penetrance estimates and improve their accuracy. Our group and others have previously deployed this approach for the full arrhythmia-associated proteins *KCNH2*,¹³ *KCNJ2*,³³ *KCNE1*,³⁴ and *CALM1-3*.³⁵ This confluence of advances in deeply phenotyped biobank-scale cohorts, high-throughput molecular biology, and variant-based penetrance promises to facilitate the clinical implementation of personalized medicine.

Limitations

The current Bayesian penetrance estimates are not conditioned on functional data, as high-throughput approaches to interrogate *RYR2* biology are currently limited. Furthermore, the affected and unaffected status was determined according to the designation of the reporting authors, since detailed phenotype data were often unavailable. As a result, some misclassification is possible, particularly considering that a subset of cases previously reported as CPVT may in fact represent CRDS. Variable penetrance is also an important consideration: some carriers develop clear CPVT while others remain unaffected despite harboring the same variant. This has been documented in founder mutation cohorts such as carriers of the G357S variant, in whom only 46% had an abnormal exercise test at the first evaluation, even after 10 tests; not every heterozygote had a CPVT phenotype.³⁶ Although we flag published CRDS variants in the Variant Browser, our current computational analysis focuses exclusively on CPVT penetrance. Given overlapping phenotypes and a much rarer frequency, it will be prudent for future massively parallel experimental work to distinguish the directionality of the variant mechanism before estimation of CRDS penetrance. Loss-of-function variants are generally more common than gain-of-function variants (ie, Brugada

Syndrome versus long-QT syndrome among *SCN5A* variants). Accordingly, there may be a larger fraction of variants for which the current *RYR2*-CPVT densities may not apply after more extensive experimental investigation.

Conclusions

Clinical genomics still strives for binary answers—pathogenic versus benign—to questions that are inherently probabilistic. Our work challenges this paradigm by treating each missense variant as a quantitative modifier of disease risk. *RYR2*-CPVT Bayesian penetrance estimates provide a pathway to build on the classic variant classification. These estimates provide structural insights into regions enhanced for highly penetrant variants and will facilitate clinical genetics and interpretation of secondary findings. Variant information is conveniently hosted at the Variant Browser for use by physicians, genetic counselors, and researchers.

ARTICLE INFORMATION

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Disclosures

None.

Supplemental Material

Tables S1–S2

Figure S1

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